

Sri Eshwar



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Thiran

CONFLUENCE OF TALENTS

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Theme : IoT

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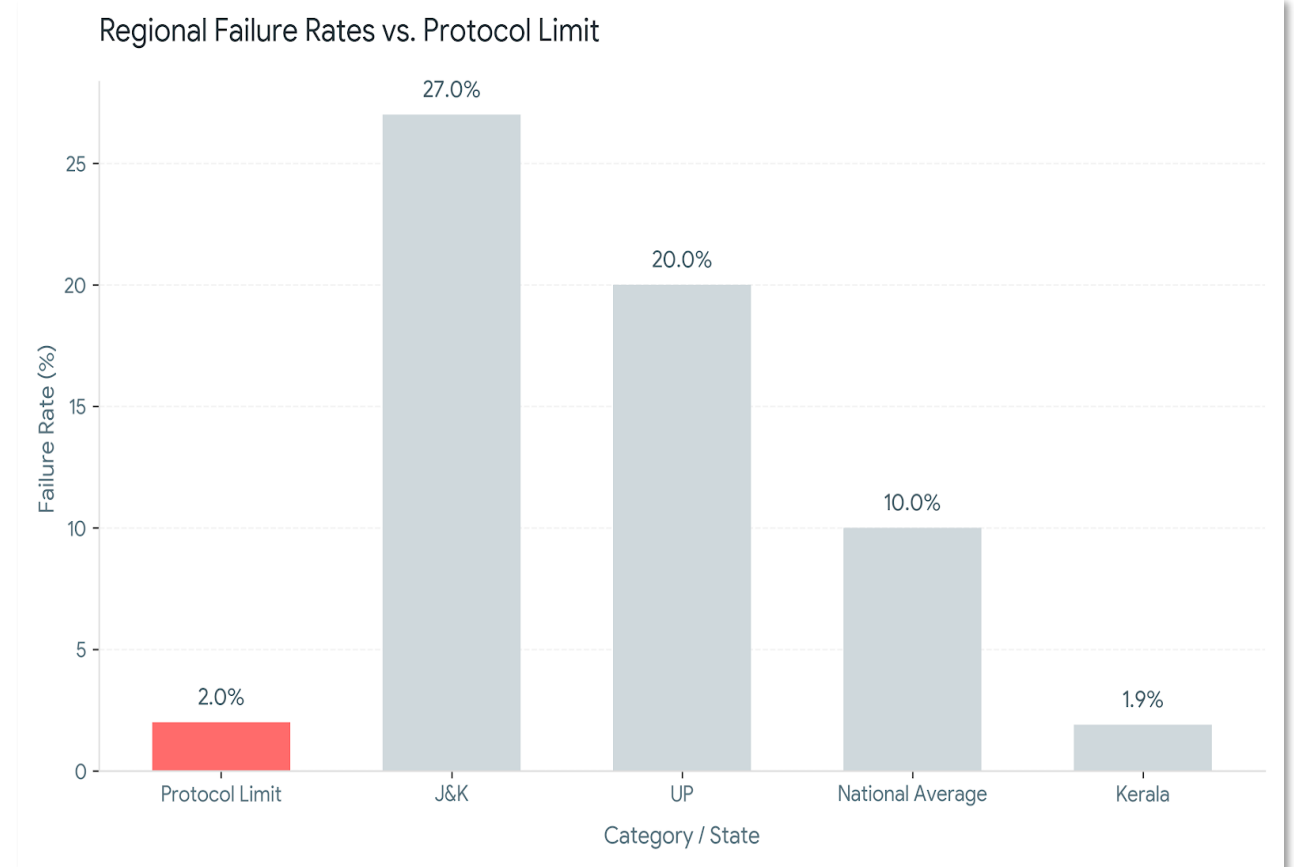
Problem Statement

MAINTENANCE TREND

- Distribution transformers are maintained on **fixed schedules** to prevent failures but not based on actual condition.

INEFFICIENCY

- Late detection of overload and thermal stress cause failures
- Unnecessary monthly inspections cause high maintenance costs and reduced reliability of power distribution networks



Proposed Solution

SOLUTION

Edge-based, context-aware transformer monitoring system

- Utilizes ESP32 + sensors for oil temperature, ambient temperature and current rating.

System enables **predictive maintenance**

- Skipping inspections when transformer is healthy
- Advancing inspections when early degradation is detected

LOAD FACTOR

LF = Potentiometer reading / Max Potentiometer reading

THERMAL STRESS INDEX

TSI = Internal Oil Temperature – Ambient Temperature

EMPHASIS ON TEMPERATURE

- Every 6-8°C rise halves insulation life
- Oil degrades → sludge → poor cooling → failure loop

Market Size

TRENDS

Indian transformer market is forecasted to grow from 3 billion USD in 2025 to 4.82 billion USD in 2031.

Utilities face:

- Rising load demand
- Increasing maintenance costs

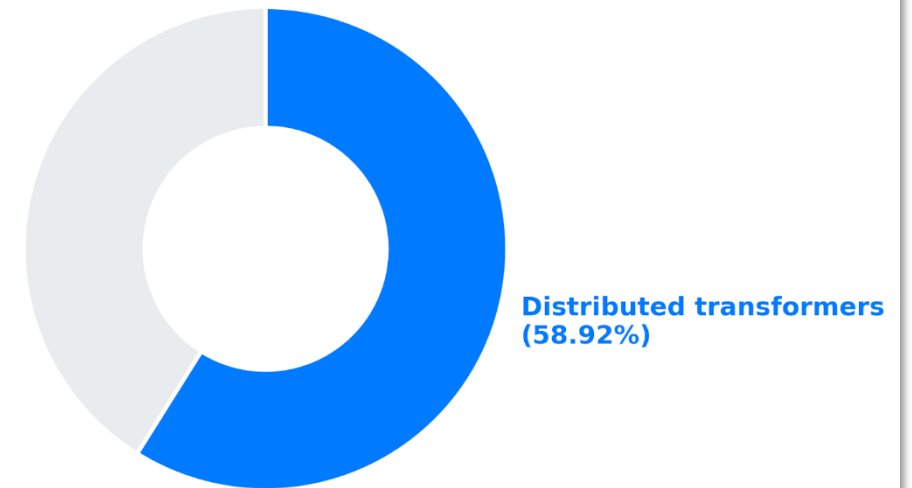
EFFECTS

Even small reduction in inspections → **large operational savings**

Scalable for:

- Urban distribution networks
- Semi-urban and rural grids

Distributed Transformers Market Share



market proportions of transformers

Uniqueness

- Moves beyond **threshold-based IoT** by using relative temperature analysis instead of fixed limits
- Actively distinguishes environmental heating from true internal overload conditions
- Introduces an **overstress duration window** to validate faults, respecting transformer thermal inertia
- Designed specifically for distribution transformers, keeping cost and complexity low

Conclusion

CONCLUSION

- Purpose-built distribution transformer monitoring system
- Lowers complexity and high costs of maintenance
- Large scale distribution networks are economically viable

FUTURE SCOPE

Can integrate:

- GSM for rural areas
- Additional sensors (humidity, oil quality, vibration)

Expandable to:

- Feeder-level analytics
- Substation dashboards

References

<https://economictimes.indiatimes.com/industry/energy/power/around-1-3-million-transformer-failures-occur-in-india-annually/articleshow/125386353.cms>

<https://www.mordorintelligence.com/industry-reports/india-transformer-market>

Thank You